

JEE MAIN REVISION MASTER: BINOMIAL THEOREM SYLLABUS REDUCED

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 160 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	General Term, Middle Term & Greatest Term	107 PYQs	Expansion Terms
Very High	Properties & Series Sum of Binomial Coefficients	56 PYQs	Coefficient Identities

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
General Term, Middle Term & Greatest Term	$(a + b)^n = \sum_{r=0}^n {}^nC_r a^{n-r} b^r$, $T_{r+1} = {}^nC_r a^{n-r} b^r$, $T_{\frac{n}{2} + 1}$ (if n is even)
Properties & Series Sum of Binomial Coefficients	$\sum_{r=0}^n {}^nC_r = 2^n$, $C_0 - C_1 + C_2 - \dots = 0$, $\sum_{r=1}^n r {}^nC_r = n 2^{n-1}$, $\sum_{r=0}^n ({}^nC_r)^2 = {}^{2n}C_n$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 General Term, Middle Term & Greatest Term (Very High Yield — 107 PYQs)

Core Principle: Expansion of $(x + a)^n$, formula for general $(r+1)^{\text{th}}$ term, finding term independent of x , and middle term determination.

■ **Caution / JEE Pitfall:** In $(a + b)^n$, total number of terms is $n + 1$. If n is odd, there are TWO middle terms: $(\frac{n+1}{2})^{\text{th}}$ and $(\frac{n+3}{2})^{\text{th}}$.

■ **Speed Trick:** In expansion of $(ax^p + bx^q)^n$, the term independent of x occurs at index $r = \frac{np}{p+q}$ (provided r is an integer).

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "Let the smallest value of $k \in \mathbb{N}$, for which the coefficient of x^3 in $(1+x)^3 + (1+x)^4 + (1+x)^5 + \dots + (1+x)^{99} + (1+kx)^{100}$, $x \dots$ " [Ans: B]
- [2026 Morning Shift] Q#1 (MCQ): "The coefficient of x^{48} in $(1+x) + 2(1+x)^2 + 3(1+x)^3 + \dots + 100(1+x)^{100}$ is equal to..." [Ans: N/A]
- [2026 Morning Shift] Q#2 (MCQ): "If the coefficients of the middle terms in the binomial expansions of $(1+\alpha x)^{26}$ and $(1-\alpha x)^{28}$, $\alpha \neq 0$, are equal, ..." [Ans: D]

3.2 Properties & Series Sum of Binomial Coefficients (Very High Yield — 56 PYQs)

Core Principle: Combinatorial identities, summing coefficient series via differentiation and integration of $(1+x)^n$.

■ **Caution / JEE Pitfall:** To find the sum of all coefficients of a polynomial expansion $P(x)$, simply substitute $x = 1$.

■ **Speed Trick:** Useful ratio identity: $\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}$.

Representative Source PYQs:

• **[2026 Evening Shift] Q#1 (MCQ):** "If

$\frac{{}^2P_3}{{}^{12}C_2} + \frac{{}^2P_5}{{}^{12}C_4} + \frac{{}^2P_7}{{}^{12}C_6} + \dots + \frac{{}^2P_{13}}{{}^{12}C_{12}} = 3^{13} - a$...

Find the value of a ." [Ans: N/A]

• **[2026 Morning Shift] Q#1 (MCQ):** "Let $S = (1/25!) + (1/3!23!) + (1/5!21!) + \dots$ up to 13 terms. If $13S = \frac{2^k n!}{k!}$, $k \in \mathbb{N}$, then $n+k$ is equal to..." [Ans: N/A]

• **[2026 Evening Shift] Q#2 (MCQ):** "If for $3 \leq r \leq 30$, $({}^{30}C_{(30-r)} + 3({}^{30}C_{(31-r)} + 3({}^{30}C_{(32-r)} + ({}^{30}C_{(33-r)} = {}^mC_r)$, then m equals :..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 4th April Morning Shift (Q#1)

Let the smallest value of $k \in \mathbb{N}$, for which the coefficient of x^3 in $(1+x)^3 + (1+x)^4 + (1+x)^5 + \dots + (1+x)^{99} + (1+kx)^{100}$, $x \neq 0$, is $(43n + (101/4))({}^{100}C_3)$ for some $n \in \mathbb{N}$, be p . Then the value of $p+n$ is :

- (A) 10
- (B) 11 [Correct]
- (C) 12
- (D) 13

Correct Answer: B

Solution Analysis: Coefficient of x^3 in the given expression $(1+x)^3 + (1+x)^4 + \dots + (1+x)^{99} + (1+kx)^{100}$ is obtained by adding the coefficients of x^3 from each term. For $(1+x)^r$, coefficient of x^3 is rC_3 . So, from $(1+x)^3 + (1+x)^4 + \dots + (1+x)^{99}$ the total coefficient of x^3 is $\sum_{r=3}^{99} {}^rC_3 = 3^{99}C_3$. Also, in $(1+kx)^{100}$, coefficient of x^3 is ${}^{100}C_3 k^3$. Hence total coefficient of x^3 is $\sum_{r=3}^{99} {}^rC_3 + {}^{100}C_3 k^3$. Now use the standard NCERT identity: $\sum_{r=3}^{99} {}^rC_3 = {}^{100}C_4$. So coefficient becomes ${}^{100}C_4 + {}^{100}C_3 k^3$.